

CMSE 381

Final Project

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Research Questions

- **RQ1:** How much information about face identity and head orientation is present in AM neural responses?

(Explored through Ridge Regression predicting firing rate from stimulus labels)

- **RQ2:** How well can identity be decoded from neural population activity?

(Explored through SVM classification)



[https://commons.wikimedia.org/wiki/File:Japanese_Snow_Monkey_\(Macaque\)_Mother_Grooms_Her_Young.jpg](https://commons.wikimedia.org/wiki/File:Japanese_Snow_Monkey_(Macaque)_Mother_Grooms_Her_Young.jpg)

a

L90

L45

F

R45

R90

ID1



ID2



What drives AM neural responses?

Identity or head orientation?

ID3

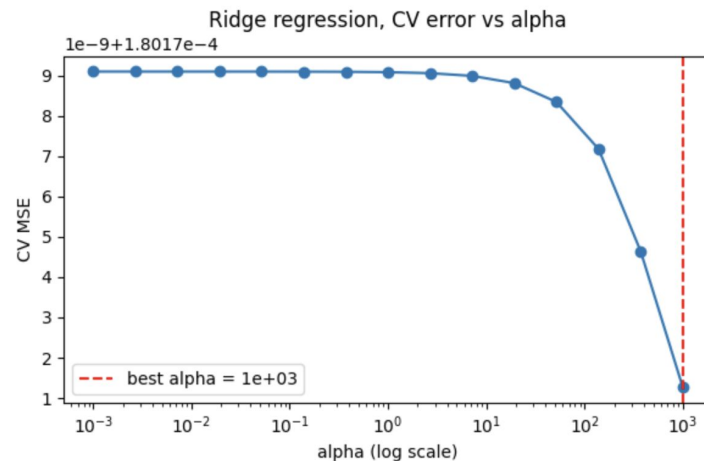
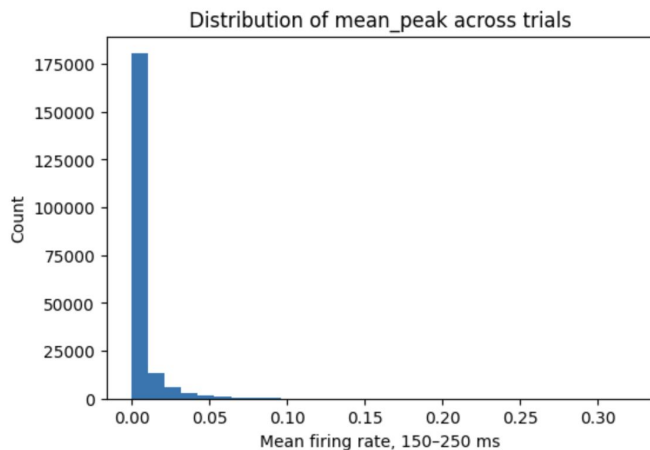


ID4



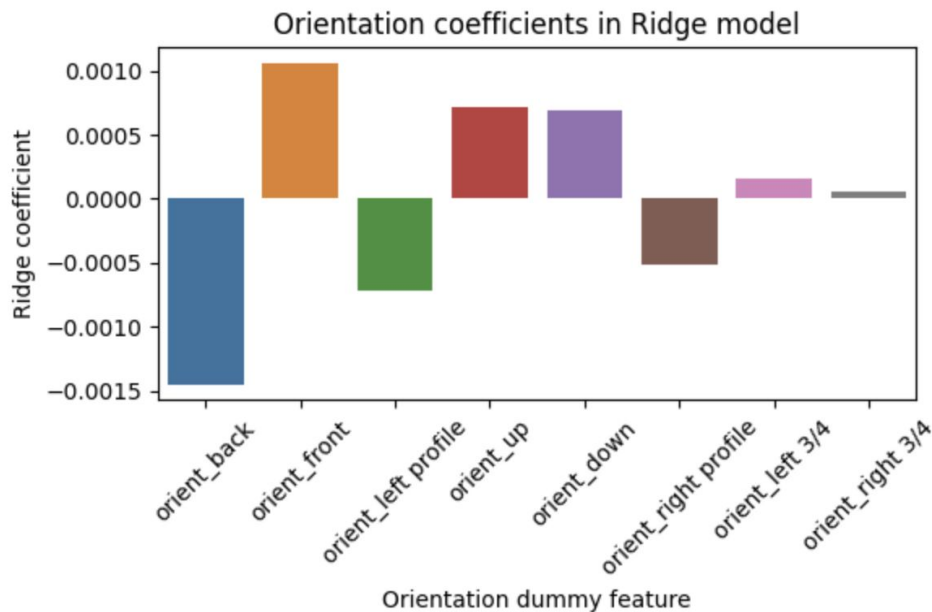
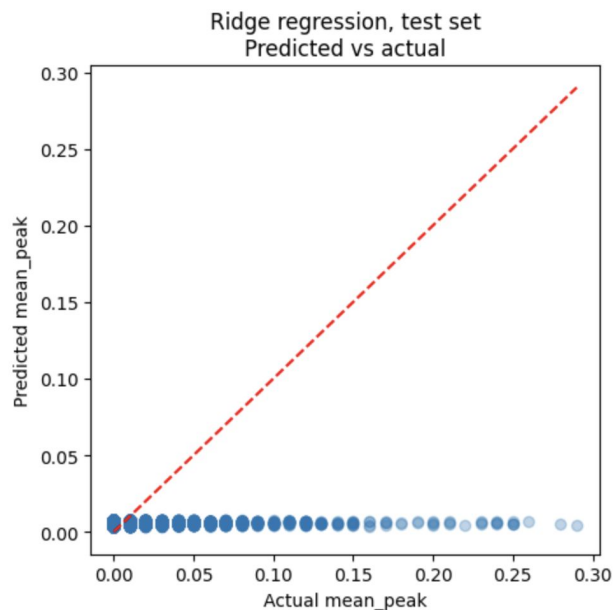
Methods Overview I

- Preprocessing: combine 193 neurons into pseudo population
- Extract 150–250 ms peak firing
- One hot encode identity + orientation
- Ridge Regression with cross validation



Key Findings I

- $R^2 < 1$ percent
- Orientation predicts slightly more than identity
- Coefficients are extremely small



Methods Overview II

- Preprocessing: Concatenate + aggregate data grouped by site, monkey, person and orientation, taking the mean of the firing rate, giving me 38,000 combinations. Subsampled 5000 combinations split between the monkeys and checked for class balance.



Methods Overview II

- Standardized + validation split subset of the data
- Conducted linear kernel default hyperparameter SVC. Received ~4% error
- Conducted hyperparameter tuning via 5-fold CV. Tested a range of C, gamma, degree values for linear, radial, and polynomial kernels.

Key Findings II

- Best kernel: poly
- Parameters: {'C': 0.1, 'degree': 4, 'gamma': 'scale', 'kernel': 'poly'}
- CV accuracy: 4.50%
- Test accuracy: 4.90%

The accuracy was better than the baseline, which was about 4.70%. However, this is still terrible.

Hence, peak firing rates may not be the best predictors of facial decoding.